PRD Genie Architecture Write-up

A governed multi-agent path from authoritative evidence to planning-ready delivery

Problem summary

NeuronForge PMs and TPMs manually reconcile product briefs, meeting transcripts, stakeholder notes, and clarifications before creating PRDs and delivery backlogs. This process is slow, inconsistent, and difficult to audit because evidence, decisions, and downstream artifacts can drift apart. PRD Genie creates a governed path from authoritative evidence to planning-ready documentation while retaining human approval, visible uncertainty, and source-to-delivery traceability.

Input-set evolution: three immutable discovery sources were followed by three human-authored clarification records dated August 7, 2026. Formal execution 11 901 consumed the resulting six-document approved packet; decision IDs and bounded supersession preserve both original evidence and current governing state.

Architecture diagram

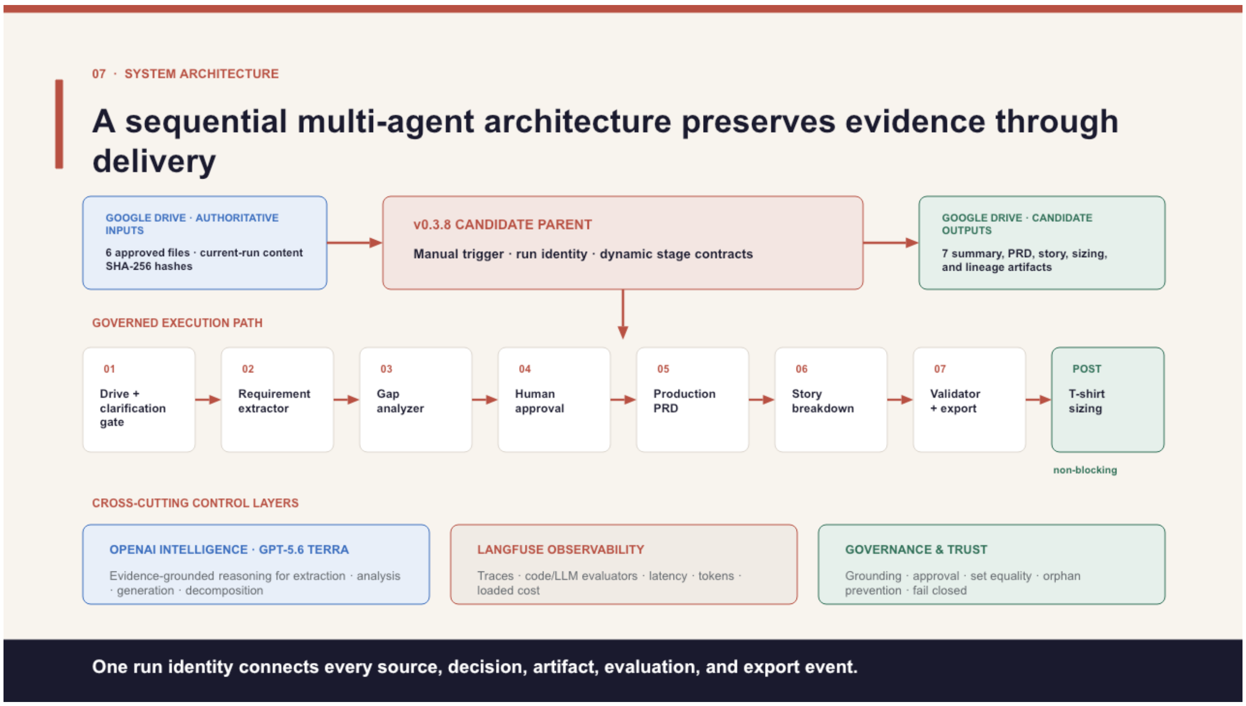

Figure 1. Approved v0.3.8 system architecture: six Google Drive inputs, a manual n8n parent, seven governed stages, non-blocking sizing, OpenAI intelligence, Langfuse observability, and fail-closed governance.

Tool selection rationale

Category	Selection and rationale
Workflow	n8n Cloud — visual parent/child orchestration, Drive integration, deterministic code nodes, and exportable JSON.
Intelligence	OpenAI GPT-5.6 Terra — constrained extraction, analysis, document generation, decomposition, and advisory sizing.
APIs / storage	Google Drive — controlled source intake and run-specific delivery with file identity, hashes, and receipts.
Observability	Langfuse — correlated traces, evaluator results, latency, token usage, and fully loaded cost.
Evidence	GitHub, Obsidian, and vjra.us — reproducibility, working documentation, and evaluator-facing access.

Orchestration pattern and justification

Sequential parent/child pipeline with conditional fail-closed branches. Each stage depends on a validated upstream contract: intake ⟶ extraction ⟶ gap analysis ⟶ human approval ⟶ PRD ⟶ stories ⟶ reconciliation/export. Clarification, rejection, validation failure, and revision stop or redirect the run. Specialized children remain independently testable while one parent run identity connects every source, decision, artifact, evaluation, and delivery event.

Why Gap Analysis immediately follows extraction: the Extractor establishes a normalized, citation-linked account of what the evidence says; the Gap Analyzer then tests sufficiency, ambiguity, contradiction, dependencies, risks, and generation readiness before approval or PRD creation. This fail-fast boundary prevents unresolved evidence from becoming polished downstream content and avoids generating artifacts that must later be discarded. Post-generation validators retain separate grounding, structure, coverage, and reconciliation duties.

Agent and control responsibilities

Stage	Input → output	Responsibility and key rules
Drive + clarification	Files → hashed evidence	Verify the current-run source set; preserve provenance; stop for missing inputs.
Requirement Extractor	Evidence → requirements	Extract only supported facts and exact values; never invent missing content.
Gap Analyzer	Requirements → gaps / route	Expose ambiguity and conflicts; route insufficient evidence to clarification.
Human Approval	Evidence + gaps → dispositions	Authorize scope and decisions; no approval means no production generation.
Production PRD	Approved evidence → PRD	Generate only approved content; preserve IDs; mark controlled unknowns.
Story Breakdown	PRD → Epics / Features / Stories	Preserve lineage and hierarchy; reject duplicates, orphans, and unsupported detail.
Validator + Export	Artifacts → authorized delivery	Reconcile sets and mappings; Agreement Gate fails closed before export.
Scope Estimator	Stories → T-shirt size estimates	Return advisory size, confidence, and guidance; remain non-blocking and non-committal.

Cost analysis

Formal baseline execution 11901: 545,467 tokens and $1.474409 fully loaded (59,922 generation tokens / $0.228024; 485,545 tokens across 41 evaluator calls / $1.246385). At one fully evaluated run per user per week: approximately **$0.21 per user per day and $6.38 per user per month**. Cost controls are code-first validation, no duplicate semantic scoring, stage-specific judges, and lighter routine monitoring after approval.

Evaluation strategy and production metrics

- Extraction completeness: percentage of stated requirements correctly captured; target 100%, with source-to-export ID reconciliation and zero orphans.
- Hallucination rate: percentage of PRD items not traceable to approved source material; target zero unsupported claims and stage hallucination ≤ 0.15.
- Format compliance: percentage of PRDs following the required ten-section template and passing schema validation; target 100%.

Method: T1–T12 ground truth; schema, exact-value, and set controls; Langfuse code, faithfulness, and hallucination evaluators; Scope Estimator reasonableness; and a fail-closed Agreement Gate. Supporting metrics include rework, time saved, latency, adoption, and loaded cost.

Testing observations and fixes

- What worked: typed contracts, explicit insufficient-information behavior, human approval, and fail-closed validation prevented unsupported output from reaching delivery.
- Hallucination and fix: T6 incorrectly treated microservices and a single-page application as contradictory. Removing the unsupported contradiction and cross-links while retaining genuine approval, deadline, and scope gaps reduced hallucination from 0.40 to 0.08; code evaluation passed.
- Stable IDs, set equality, bidirectional lineage, and final reconciliation closed duplicate, dropped-mapping, and orphan risks.
- Removing duplicate semantic scoring reduced evaluator overhead without weakening coverage.

ACCEPTED RESULT Execution 11 901 completed in 2m 33.201s, authorized release, reconciled 145/145 citation dispositions with zero orphaned governed records, delivered seven validated artifacts, and sized 11/11 stories. Execution 11 958 remains supplementary demonstration evidence.